Memory-Defined Phase Maps for Dissipative Quantum Reservoirs

Anonymous QCE Submission

***Abstract*—Quantum reservoir computing performs temporal machine learning by training only a linear readout on the measured outputs of a fixed quantum system, leaving the quantum dynamics itself untrained. This makes the approach attractive for near-term hardware, but it sharpens a question that has remained largely unresolved: which device settings render the fixed dynamics useful rather than merely high-dimensional. We address this question for a stateful dissipative gate-model reservoir and demonstrate that strong performance does not occupy arbitrary coordinates of the control space. Instead, it concentrates in a well-defined operating region governed by the interplay of input drive, recurrent coupling, and controlled dissipation. This region is transferable: an operating band identified on a subset of tasks predicts effective settings on held-out tasks and held-out reservoir initializations alike. Mechanism ablations establish that the region is causal rather than incidental, and an inexpensive memory-based diagnostic recovers it prior to any task-specific evaluation. The result reframes reservoir selection as the identification of a transferable, memory-preserving regime rather than the pursuit of a single benchmark optimum.**

***Index Terms*—quantum reservoir computing, dissipative dynamics, memory capacity, phase diagrams, operating regimes**

I. Introduction

Reservoir computing rests on an unusually economical idea: a fixed dynamical system, left untrained, transforms an input signal into a high-dimensional trace of its recent history, and learning reduces to fitting a single linear readout on top of that trace [1], [2]. The dynamics are never optimized, only observed. This economy is precisely what makes the approach appealing for quantum hardware, where the native evolution of a system can serve as the reservoir and only the classical readout must be trained. Yet the same economy relocates the entire burden of the method onto the fixed dynamics, and so onto a question that is easy to state and surprisingly hard to answer: what makes those dynamics useful?

Classical reservoir theory has long since refused the obvious answer. Usefulness is not maximal expressivity. A reservoir earns its keep by holding a balance—retaining recent input, forgetting old input in a controlled way, and exposing nonlinear functions of that history to the readout [3], [4], [5]. A quantum reservoir inherits this lesson but complicates it, because the three ingredients are no longer free dials. They are downstream consequences of distinct physical resources: how strongly the input perturbs the carried state, how coherently the system mixes that state across its parts, and how it is allowed to dissipate [4], [6]. A large Hilbert space supplies dimension, but dimension alone preserves no memory; drive that is too strong overwrites history, and dynamics with no dissipation never forget in the way fading memory requires.

We take this seriously as an organizing principle rather than a caveat. For a stateful dissipative gate-model reservoir, we treat the space of these three resources as the proper setting in which to ask when the reservoir works, and we map it. The picture that emerges is not a single tuned operating point but a connected region—gentle input, nonmaximal coupling, controlled damping—in which good performance concentrates and from which it does not stray arbitrarily across tasks. The contribution of this paper is that region itself: its existence, its transfer across tasks and reservoir initializations, the mechanisms that sustain it, and a memory-based diagnostic that locates it before any task-specific search.

II. Protocol

The reservoir carries a density matrix from one input time step to the next. For scalar input u_t,

$$\rho_{t,0} = U_{\mathrm{in}}(u_t)\rho_{t-1,L}U_{\mathrm{in}}^{\dagger}(u_t), \qquad U_{\mathrm{in}}(u_t) = \bigotimes_i R_y(\beta u_t). \tag{1}$$

Each virtual layer applies frozen local rotations, a ZZ ring, and a product noise channel:

$$\begin{aligned} \rho_{t,\ell} &= \mathcal{D}_{\gamma}^{\otimes n}\left[U_{\lambda}U_{\mathrm{mix}}\rho_{t,\ell-1}U_{\mathrm{mix}}^{\dagger}U_{\lambda}^{\dagger}\right], \\ U_{\lambda} &= \prod_{(i,j)\in E_{\mathrm{ring}}} e^{-i\lambda Z_i Z_j}. \end{aligned} \tag{2}$$

The main phase maps use $n = 4$, two layers, ring topology, uniform input, amplitude damping, the $Z + ZZ$ QRC16 readout, and ridge-only training. Hyperparameters are never selected by holdout data.

For task j and seed s, let $r_{j,s}(\theta)$ be the validation percentile rank of $\theta = (\beta, \lambda, \gamma)$, with lower rank better. A transferable operating band is

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \le p] \ge q\}. \tag{3}$$

This frequency-level definition is deliberately weaker than a universal optimum. It asks whether a region repeatedly behaves well, not whether the same coordinate wins every task.

III. Evidence

Phase-map transfer. Fig. 1 shows the core pattern: low validation rank is concentrated, not randomly scattered. The primary band $B_{20,0.7}$ has 4 connected points, all at $\gamma = 0.12$. When one task is withheld while defining the band from the remaining tasks, the held-out task-seed mean validation rank

TABLE I

QRC-ONLY AUDIT TRAIL. HOLDOUT DATA ARE NOT USED TO DEFINE THE OPERATING REGIME.

Item	Value
Tasks and splits	Mackey–Glass, Lorenz, NARMA10: washout/train/validation/holdout $= 300/1600/400/400$; annual Sunspots: $20/150/50/60$. Inputs are scaled to $[-1, 1]$ and targets standardized.
Seeds and ridge	Reservoir seeds 42–51; ridge grid $\{10^{-8}, 10^{-7}, \ldots, 10^{3}\}$ selected on validation.
Phase grid	QRC16 coarse grid over $\beta/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\lambda/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, $\gamma = \{0, 0.05, 0.12, 0.22, 0.30\}$.
Primary band	$B_{20,0.7}$ contains 4 connected points; the medoid is $(\beta/\pi, \lambda/\pi, \gamma) = (0.04, 0.12, 0.12)$. Relaxing to $B_{20,0.6}$ gives 14 points and $B_{30,0.7}$ gives 46 points.
Stress tests	Focused QRC-only ablation maps at $\gamma = 0.12$ compare base amplitude damping against $\gamma = 0$, dephasing, depolarizing, no mixer, Rx-only mixer, and ZZ-only mixer. The readout, tasks, seeds, and ridge protocol are unchanged.
Diagnostics	Memory capacity and intrinsic processing capacities are computed from simulator trajectories and compared to validation rank and top-decile screening retention.

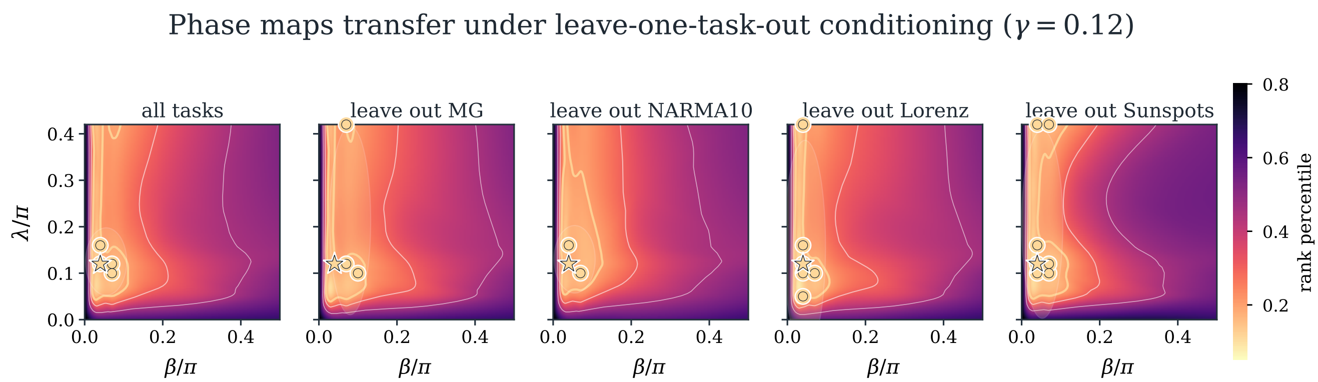

Fig. 1. Validation phase maps at the fixed $\gamma = 0.12$ slice. Bright regions are low validation percentile rank. Gold circles mark points that appear in the top-20% for at least 70% of the displayed task-seed replicates; the star is the medoid of the primary band. The same low-input, nonmaximal-coupling region persists under leave-one-task-out conditioning.

TABLE II

HOLDOUT PERFORMANCE AFTER VALIDATION-ONLY BAND SELECTION. MEANS ARE OVER TEN SEEDS PER TASK; "ALL" AVERAGES THE 40 TASK-SEED ROWS.

Task	Band NMSE	Medoid NMSE	Band rank
Mackey–Glass	8.77e-05	8.19e-05	0.105
Lorenz	5.28e-05	7.36e-05	0.169
NARMA10	0.561	0.520	0.172
Sunspots	0.194	0.179	0.177
All	0.189	0.175	0.156

is 0.240 with bootstrap CI $[0.215, 0.265]$. When one seed is withheld, the corresponding mean rank is 0.180 with CI $[0.154, 0.209]$. The band is therefore a top-quartile search prior that transfers across both task identity and reservoir initialization.

Final holdout audit. After $B_{20,0.7}$ is fixed by validation ranks, the selected band is evaluated on the held-out split. Table II reports the mean over the validation-selected band and the fixed medoid; no coordinate is chosen by holdout NMSE. The aggregate band holdout rank is 0.156, so the operating-regime prior remains top-quartile on unseen data.

Mechanism stress tests. The operating band is not just a plotting artifact. Figure 2(a) shows the validation-defined frequency map for the primary band, and Fig. 2(b–d) shows where rank is lost when mechanisms are removed. At the central slice, the projected base band has 4 points. Removing the controlled damping mechanism gives retention 0.00; removing recurrent mixing gives retention 0.00; replacing amplitude damping by dephasing gives retention 0.00. In every focused ablation, the base band ceases to be a robust top-ranked region under the same validation-rank criterion. The result supports the mechanism story: useful dynamics need input that is not too strong, recurrent mixing, and damping that preserves usable memory rather than merely adding noise.

Variant	Band pts.	Retention	Rank on base band
base AD+RxZZ	4	1.00	0.256
$\gamma = 0$	0	0.00	0.465
dephasing	0	0.00	0.442
depolarizing	0	0.00	0.526
no mixer	0	0.00	0.403
Rx only	0	0.00	0.521
ZZ only	0	0.00	0.503

Diagnostics. Memory-based diagnostics identify the same useful region before task-specific holdout evaluation. On the current simulator-backed diagnostic rerun, MC has Spearman $\rho_s = -0.85$ against validation rank; $\mathrm{IPC}_{\mathrm{mem}}$, $\mathrm{IPC}_{\mathrm{tot}}$, and $\mathrm{IPC}_{\mathrm{nonlin}}$ give -0.84, -0.90, and -0.80. Feature variation and effective rank are weakly positive, 0.16 and 0.15, so raw diversity alone is not the mechanism.

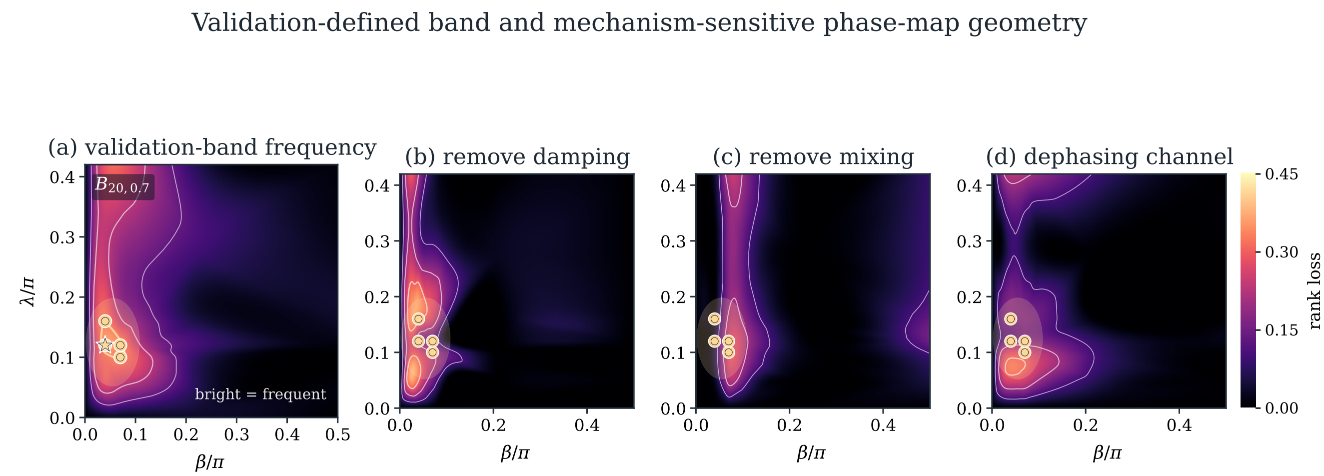

Fig. 2. QRC-only phase-map evidence for the operating-regime hypothesis. (a) The top-20 frequency map visualizes the validation-defined $B_{20,0.7}$ band at $\gamma = 0.12$: bright coordinates repeatedly rank in the top validation quintile across task-seed replicates. (b–d) Ablation rank-loss maps show how much the mean validation rank degrades when controlled damping is removed, recurrent mixing is removed, or amplitude damping is replaced by dephasing. The highlighted base-band projection lies where these mechanism removals cause rank loss.

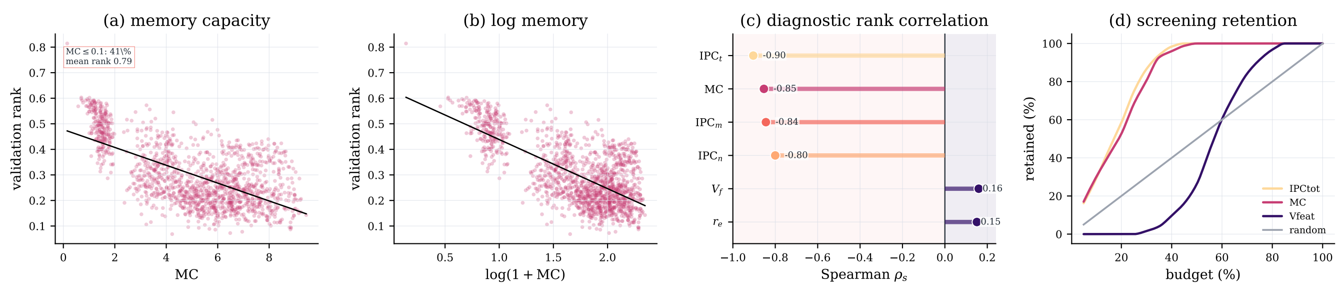

Fig. 3. Diagnostic audit for the QRC phase-map claim. Raw and log-scaled memory capacity are shown for active-memory configurations; the zero/near-zero memory cluster is aggregated in the panel annotation. Intrinsic processing-capacity diagnostics give the strongest rank correlations, and screening by memory/IPC recovers useful reservoirs earlier than random or diversity-only screens.

IV. Discussion and Scope

The evidence supports a universal operating-regime hypothesis, not a theorem. The tested tasks differ in chaotic smooth dynamics, nonlinear memory, and real annual observations, and the same frequency-defined phase region remains useful across seeds and leave-one-task-out splits. The statement should still be read within the modeled family: four-qubit stateful gate-model QRCs, scalar time-series prediction, exact simulated expectation values, ridge readouts, and the specified grid.

The scope is intentionally narrower than a hardware or advantage claim. Finite-shot sampling, measurement-basis scheduling, calibration drift, and device noise are not modeled. Likewise, the phase map does not say that every task has the same optimum. It says that useful QRC learning should first be searched in a memory-preserving band, and that leaving this band by removing damping or recurrent mixing destroys the robust phase-map signature.

V. Conclusion

The usefulness of a dissipative quantum reservoir is not a property of its size but of where it is operated. Across tasks and reservoir initializations, strong performance does not wander the control space; it gathers in a single memory-preserving region where input drive, recurrent coupling, and dissipation hold one another in balance. That region transfers, it survives the removal of the very mechanisms that define it only by collapsing, and it can be found in advance by a diagnostic that asks nothing about the task—only whether the reservoir remembers.

The lesson is deflationary in the most productive sense. The resource that makes this quantum reservoir work is not a quantum mystery but the same fading-memory-and-nonlinearity balance that classical reservoir theory identified long ago [3], [5], realized here through physical channels rather than chosen weights. Read this way, the contribution is methodological before it is physical: it reframes reservoir selection as the task of locating a transferable operating regime, and it offers memory, rather than raw feature diversity, as the compass that points there.

What remains is the harder half. These results are established in simulation, on a small family of architectures, with measurement treated as free; finite sampling, calibration drift, device noise, and the true cost of reading out many observables all lie outside the present claim. The natural next step is not a larger benchmark but a more honest one—to ask whether the same memory-defined regime persists once the reservoir must be measured, calibrated, and run on hardware that does not forget on command. We expect it to, and we have given the map for finding out.

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